

Association Between Body Mass Index and CKD in Apparently Healthy Men

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● **Background:** Overweight and obesity are well-established risk factors for cardiovascular disease and decline in kidney function in individuals with existing chronic kidney disease (CKD). Conversely, their association with the development of CKD is less clear. **Methods:** We evaluated the association between body mass index (BMI) and risk for CKD in a cohort of 11,104 initially healthy men who participated in the Physicians' Health Study and provided a blood sample after 14 years. BMI was calculated from self-reported weight and height. We estimated glomerular filtration rate (GFR) by using the abbreviated equation from the Modification of Diet in Renal Disease Study and defined CKD as GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²). **Results:** After an average 14-year follow-up, 1,377 participants (12.4%) had a GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²). Higher baseline BMI was associated consistently with increased risk for CKD. Compared with participants in the lowest BMI quintile (<22.7 kg/m²), those in the highest quintile (>26.6 kg/m²) had an odds ratio (OR) of 1.45 (95% confidence interval [CI], 1.19 to 1.76; *P* trend <0.001) after adjusting for potential confounders. We found similar associations by using different categories of BMI. Compared with men who remained within a $\pm 5\%$ range of their baseline BMI, those who reported a BMI increase greater than 10% had a significant increase in risk for CKD (OR, 1.27; 95% CI, 1.06 to 1.53). **Conclusion:** In this large cohort of initially healthy men, BMI was associated significantly with increased risk for CKD after 14 years. Strategies to decrease CKD risk might include prevention of overweight and obesity. *Am J Kidney Dis* 46:871-880.

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KIDNEY FAILURE is prevalent in the United States, with increasing incidence, poor outcomes, and high costs of treatment.¹⁻⁵ Approximately 20 million adults in the United States have earlier stages of chronic kidney disease (CKD), and these individuals bear a disproportionate burden of cardiovascular disease and related mortality.^{3,4} CKD is defined by the National Kidney Foundation as a persistent decrease in glomerular filtration rate (GFR) to less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²) or the presence of markers of kidney damage, such as albuminuria.⁶ Adverse outcomes associated with CKD include progressive loss of kidney function, greater rates of hospitalization, cardiovascular and infectious diseases, malnutrition, and all-cause mortality.^{2-4,7,8} Estimated costs for the treatment of patients with kidney failure alone exceed \$22 billion annually.^{2,5}

Although prevention relies in part on modification of underlying risk, few factors have been identified in the development and progression of CKD. Established precursors include diabetes mellitus,^{6,9} hypertension,⁶ and cholesterol level abnormalities.^{10,11} Additional risk factors that have been variably associated include physical inactivity, smoking, and alcohol consumption.¹²⁻¹⁴ Although some studies suggested that CKD may

share other risk factors traditionally related to cardiovascular disease,^{10,13-17} the association between excess weight and CKD remains less clear.^{12,15,18-20}

Excess weight is recognized as a primary risk factor for cardiovascular disease²¹⁻²⁴ and the precursors of CKD, including diabetes mellitus²⁵

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and hypertension.²⁶ However, it is unclear whether body mass index (BMI) is associated with the development of CKD independent of its relation with these precursors. We thus examined the association of BMI and change in BMI with CKD in a prospective cohort of more than 11,000 initially healthy US male physicians.

METHODS

Study subjects were participants in the Physicians' Health Study (PHS), a completed randomized trial of aspirin or beta carotene in the primary prevention of cardiovascular disease and cancer. The design, methods, and results of the PHS have been described in detail elsewhere.^{27,28} The study included 22,071 apparently healthy male physicians without a history of cardiovascular disease, cancer (except nonmelanoma skin cancer), current liver disease or decreased kidney function (defined as renal failure or insufficiency), or other major illnesses.

Baseline and follow-up information were self-reported and collected through mailed questionnaires every 6 months for the first year and annually thereafter. Data gathered included information on participant demographics; personal characteristics, including weight and height; medical history; health behaviors; compliance with medications; and newly diagnosed conditions.

Blood Collection and Analysis

The method of blood collection has been published in detail previously.^{29,30} Briefly, participants were invited to provide an EDTA blood sample during follow-up in 1996. Of 11,360 blood samples received, creatinine could be measured in 11,104.

Creatinine was analyzed in Oxford, England, by using an automated Jaffé rate method on a Synchron LX20 autoanalyzer (Beckman Coulter, Fullerton, CA) for quantification of creatinine. Plasma creatinine is stable in chilled next-day whole-blood samples preserved with EDTA.^{31,32} To assess quality control, blinded duplicate splits were submitted; the coefficient of variation for creatinine concentration in these blinded split samples was 7.1%. The difference in means between study samples and repeated quality control samples was 0.018 mg/dL (1.6 μ mol/L) $\pm$ 0.67 (SD) mg/dL. Intra-batch coefficients of variation on internal quality-control runs were 1.4% to 3.6%.

Assessment of BMI

BMI was calculated from self-reported weight in kilograms divided by height in meters squared. Baseline weight and height were available for 11,104 participants who also provided a blood sample in 1996. We evaluated BMI at baseline as: (1) quintiles, (2) a continuous term, and (3) according to World Health Organization (WHO) categories (normal, <25.0; overweight, 25.0 to 29.9; and obese, $\geq$ 30.0 kg/m²). In secondary analyses, we also evaluated changes in BMI between baseline and 14-year follow-up for the 10,906 men with available BMI values at both times.

Outcomes

The primary outcome of interest is CKD at 14-year follow-up, defined as (1) GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²) or (2) creatinine level of 1.5 mg/dL or greater ($\geq$ 133 μ mol/L). GFR was estimated by using the abbreviated prediction equation developed from the Modification of Diet in Renal Disease (MDRD) Study,³³⁻³⁵ as recommended by the National Kidney Foundation⁶:

$$\begin{aligned} \text{GFR (mL/min/1.73 m}^2 \text{ body surface area)} \\ = [186 \times (\text{creatinine [mg/dL]})^{-1.154} \times (\text{age})^{-0.203} \\ \times 1.212 \text{ (if black)}] \end{aligned}$$

In a sensitivity analysis, we also examined the association of BMI with lower GFR (<50 mL/min/1.73 m² [$<$ 0.83 mL/s/1.73 m²]). We did not consider creatinine clearance estimation based on Cockcroft and Gault³⁶ because weight is part of this equation.

Statistical Analysis

All analyses were performed using SAS software, version 8.2 (SAS Institute, Cary, NC). We compared baseline participant characteristics according to BMI categories by chi-square tests for categorical variables and analysis of variance for continuous variables. We used logistic regression to examine the association between BMI and CKD.

We calculated age- and multivariable-adjusted odds ratios (ORs) and corresponding 95% confidence intervals (CIs). We considered 2 multivariable-adjusted models: the first controlled for the potential confounders age in 5-year increments (40 to 44, 45 to 49, 50 to 54, 55 to 59, 60 to 64, 65 to 69, and $\geq$ 70 years), smoking status at baseline (never, past, and current), physical activity at baseline (none, 1 time/mo to 4 times/wk, and $\geq$ 5 times/wk), alcohol consumption ($\geq$ 1 drink/d, 2 to 6 drinks/wk, and $\leq$ 1 drink/wk), and parental history of myocardial infarction before the age of 60 years. The second multivariable regression model additionally controlled for factors considered to be potential biological mediators of the association between BMI and CKD: history of hypertension at baseline or during follow-up (defined as self-reported systolic blood pressure $\geq$ 140 mm Hg, diastolic blood pressure $\geq$ 90 mm Hg, or antihypertensive medication use regardless of blood pressure), history of diabetes at baseline or during follow-up, history of high total cholesterol level at baseline or follow-up (defined as $\geq$ 240 mg/dL [$\geq$ 6.21 mmol/L] or history of past or current use of cholesterol-lowering medication), and the development of cardiovascular disease (myocardial infarction, stroke, angina, coronary artery bypass graft, or percutaneous coronary angioplasty) during follow-up.

We further examined whether the association between BMI and risk for CKD was modified by baseline hypertension status, diabetes, age (< or $\geq$ 60 years), or physical activity (exercise < or $\geq$ 2 times/wk) in separate models. To test for effect modification, we used the likelihood ratio test contrasting age-adjusted models with and without interaction terms for baseline BMI and these variables. In secondary analyses, we excluded participants with a BMI of 30

Table 1. Characteristics of Participants According to Baseline BMI

Characteristics	BMI (kg/m ²)			P*
	<25.0 (n = 6,502)	25-29.9 (n = 4,204)	≥30.0 (n = 398)	
Age at baseline (y)	52.4 ± 8.7	53.6 ± 8.3	52.7 ± 7.7	<0.001
Hypertension† (%)				
Never	63.7	48.6	34.7	<0.001
Baseline	16.7	25.9	39.4	
Follow-up	19.6	25.5	25.9	
Diabetes mellitus (%)				
Never	96.4	91.7	79.4	<0.001
Baseline	1.1	1.8	2.8	
Follow-up	2.5	6.5	17.8	
Cardiovascular events‡ during follow-up (%)	13.2	18.3	20.1	<0.001
Smoking at baseline (%)				
Never	54.4	49.0	42.6	<0.001
Past	37.5	41.9	46.9	
Current	8.1	9.1	10.6	
Alcohol intake at baseline (%)				
≤1 drink/wk	36.8	40.5	50.0	<0.001
2-6 drinks/wk	37.3	36.3	32.6	
≥1 drink/d	25.9	23.2	17.4	
Physical activity at baseline (%)				
Rarely/never	10.5	13.5	16.0	<0.001
1 time/mo to 4 times/wk	70.5	73.6	74.4	
≥5 times/wk	19.0	13.0	9.6	
Cholesterol ≥ 240 mg/dL or treatment (%)				
Never	74.8	72.8	76.4	0.05
Baseline	10.4	11.7	11.8	
Follow-up	14.8	15.5	11.8	
Parental history of myocardial infarction (%)	9.2	10.0	12.6	0.05

NOTE. N = 11,104. Values expressed as mean ± SD or percent. To convert cholesterol in mg/dL to mmol/L, multiply by 0.0259.

*P from analysis of variance for continuous and chi-square test for trend for categorical variables.

†Defined as self-reported systolic blood pressure of 140 mm Hg or greater or diastolic blood pressure of 90 mm Hg or greater or antihypertensive medication use regardless of blood pressure.

‡Defined as self-reported myocardial infarction, stroke, angina, coronary artery bypass graft, or percutaneous coronary angioplasty.

kg/m² or greater at baseline, given potential misclassification of GFR at the greatest BMI levels.

RESULTS

Characteristics of the 11,104 participants with no prior history of kidney disease are listed according to WHO BMI categories in Table 1. As expected, overweight (BMI, 25 to 29.9 kg/m²) and obese (BMI ≥ 30 kg/m²) participants were more likely to report hypertension, diabetes, or subsequent cardiovascular events. They also were more likely to smoke, were less physically active, and reported less alcohol consumption at baseline.

Follow-up creatinine levels were obtained an average of 14 years after study enrollment. Overall, creatinine and GFR measures appeared similar among baseline BMI categories, although statistically significant differences were noted in unadjusted comparisons (Table 2). A total of 1,377 participants (12.4%) had a GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²), and 485 participants (4.4%) had a creatinine level of 1.5 mg/dL or greater (≥133 μmol/L).

In age- and multivariable-adjusted analyses, greater baseline BMI was associated significantly with increased risk for CKD at 14-year follow-up (Table 3). Age-adjusted estimated risk for CKD (GFR < 60 mL/min/1.73 m² [<1

Table 2. Unadjusted Kidney Function at 14-Year Follow-Up According to Baseline BMI

	BMI Quintile (kg/m ²)					P*
	<22.7 (n = 2,202)	22.7-23.7 (n = 2,277)	23.8-25.0 (n = 2,155)	25.1-26.6 (n = 2,250)	>26.6 (n = 2,220)	
Creatinine (mg/dL)	1.01 ± 0.23	1.04 ± 0.29	1.05 ± 0.32	1.06 ± 0.32	1.06 ± 0.42	<0.001
Creatinine ≥1.5 mg/dL (%)	3.2	4.5	4.3	4.4	5.5	0.006
GFR† (mL/min/1.73 m ²)	83.1 ± 20.4	80.4 ± 19.0	79.4 ± 18.3	78.9 ± 18.6	79.7 ± 20.6	<0.001
GFR† <60 mL/min/1.73 m ² (%)	9.5	11.9	12.3	14.2	14.0	<0.001

NOTE. Values expressed as mean ± SD or percent unless otherwise noted. To convert creatinine in mg/dL to μmol/L, multiply by 88.4; GFR in mL/min/1.73 m² to mL/s/1.73 m², multiply by 0.01667.

*P from analysis of variance for continuous and chi-square test for trend for categorical variables.

†Estimated by using the MDRD equation.

mL/s/1.73 m²]) generally increased across BMI quintiles relative to the referent group (BMI < 22.7 kg/m²).

Higher BMI remained independently associated with risk for CKD after adjustment for the potential confounders age, smoking status, alcohol consumption, exercise, and family history of premature myocardial infarction. The multivariable-adjusted OR of GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²) for men in the greatest BMI quintile (>26.6 kg/m²) was 1.45 (95% CI, 1.19 to 1.76; P for trend <0.001) compared with those in the lowest BMI quintile (<22.7 kg/m²). Examining baseline BMI as a continuous term, each 1-unit increase in BMI

was associated with a 5% (95% CI, 3% to 7%) increase in risk for CKD. We found similar associations evaluating BMI according to WHO categories. The confounder-adjusted OR for GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²) was 1.25 (95% CI, 1.11 to 1.41) for men who were overweight (BMI, 25.0 to 29.9 kg/m²) and 1.32 (95% CI, 0.96 to 1.80; P for trend <0.001) for men who were obese (BMI ≥ 30.0 kg/m²) compared with men with normal BMI (<25.0 kg/m²).

Results for a more stringent definition of CKD were similar, but somewhat stronger in magnitude. Adjusting for potential confounders, the OR for GFR less than 50 mL/min/1.73 m² (<0.83

Table 3. Age- and Multivariable-Adjusted ORs and 95% CIs for CKD at 14-Year Follow-Up According to Baseline BMI

Baseline BMI (kg/m ²)	No. of Patients	Age-Adjusted OR (95% CI)	Model 1* OR (95% CI)	Model 2† OR (95% CI)
Continuous		1.05 (1.02-1.07)	1.05 (1.03-1.07)	1.03 (1.01-1.05)
Quintiles				
1 (<22.7)	2,202	1.00	1.00	1.00
2 (22.7-23.7)	2,277	1.19 (0.98-1.45)	1.17 (0.96-1.43)	1.13 (0.92-1.38)
3 (23.8-25.0)	2,155	1.23 (1.01-1.50)	1.24 (1.02-1.52)	1.15 (0.94-1.41)
4 (25.1-26.6)	2,250	1.44 (1.19-1.75)	1.44 (1.19-1.75)	1.32 (1.09-1.61)
5 (>26.6)	2,220	1.43 (1.18-1.73)	1.45 (1.19-1.76)	1.26 (1.03-1.54)
P for trend		<0.001	<0.001	0.007

NOTE. CKD defined as a GFR less than 60 mL/min/1.73 m² (<1.00 mL/s/1.73 m²).

*Model 1 adjusted for age at baseline (40 to 44, 45 to 49, 50 to 54, 55 to 59, 60 to 64, 65 to 69, and ≥70 years), smoking at baseline (never, past, current), alcohol consumption at baseline (≥1 drink/d, 2 to 6 drinks/wk, ≤1 drink/wk), exercise (none, 1 time/mo to 4 times/wk, ≥5 times/wk), and parental history of myocardial infarction before the age of 60 years.

†Model 2 adjusted for all variables in model 1 plus diabetes (history at baseline or during follow-up), hypertension at baseline or follow-up (defined as self-reported systolic blood pressure ≥140 mm Hg or diastolic blood pressure ≥90 mm Hg or antihypertensive medication use regardless of blood pressure), elevated cholesterol level at baseline or during follow-up (defined as ≥240 mg/dL [6.21 mmol/L] or history of past or current use of cholesterol-lowering medication), and development of cardiovascular disease during follow-up (defined as self-reported myocardial infarction, stroke, angina, coronary artery bypass graft, or percutaneous coronary angioplasty).

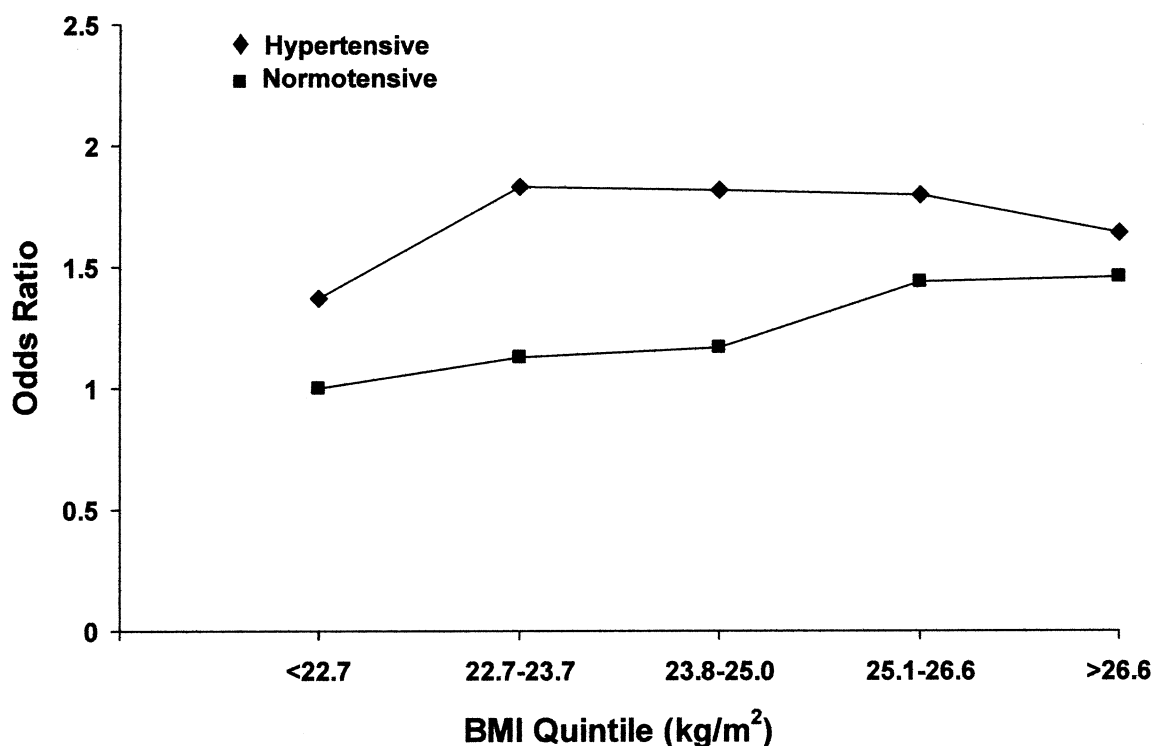


Fig 1. Age-adjusted ORs for decreased GFR (<60 mL/min/1.73 m² [<1.00 mL/s/1.73 m²]) at 14-year follow-up according to BMI quintile and hypertension status at baseline. Referent group (OR, 1.0): normotensive with BMI less than 22.7 kg/m² (P for trend: hypertensive, $P = 0.70$; normotensive, $P < 0.001$).

mL/s/1.73 m²) was 1.62 (95% CI, 1.21 to 2.18; P for trend = 0.003) for those in the greatest baseline BMI quintile compared with those in the lowest quintile. Similarly, individuals in the greatest BMI quintile also had a 1.6-fold increased risk (OR, 1.63; 95% CI, 1.20 to 2.22; P for trend = 0.01) for having a follow-up creatinine level of 1.5 mg/dL or greater (≥ 133 μ mol/L) compared with those in the lowest quintile (data not shown).

Additional adjustment for potential biological mediators (hypertension, diabetes, high cholesterol level, and development of cardiovascular disease) only partly attenuated the effect estimates of the association between BMI and risk for kidney disease (Table 3). Compared with those in the lowest BMI quintile, men in the highest BMI quintile had an OR of 1.26 (95% CI, 1.03 to 1.54; P for trend = 0.007) for developing a GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²). We found similar associations between baseline BMI and risk for kidney disease after excluding those with potential biological mediators

at baseline or follow-up ($n = 6,743$). Men in the highest BMI quintile had an OR of 1.85 (95% CI, 1.28 to 2.69; P for trend = 0.001) for developing a GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²) compared with those in the lowest BMI quintile (data not shown).

We evaluated whether hypertension, diabetes, age, or physical activity modified the association between BMI and CKD in age-adjusted analyses. Risk for CKD (GFR < 60 mL/min/1.73 m² [<1 mL/s/1.73 m²]) was consistently greatest for individuals with a history of hypertension at baseline across all BMI levels compared with normotensive individuals (Fig 1). Although risk for CKD was related most strongly to BMI in individuals who were normotensive at baseline, hypertension status did not statistically significantly modify the association between baseline BMI and CKD ($P = 0.34$). BMI was significantly related to risk for CKD in both physically active and inactive men (P for trend = 0.004 and 0.007, respectively). Physically active and inactive individuals generally had a similar risk for

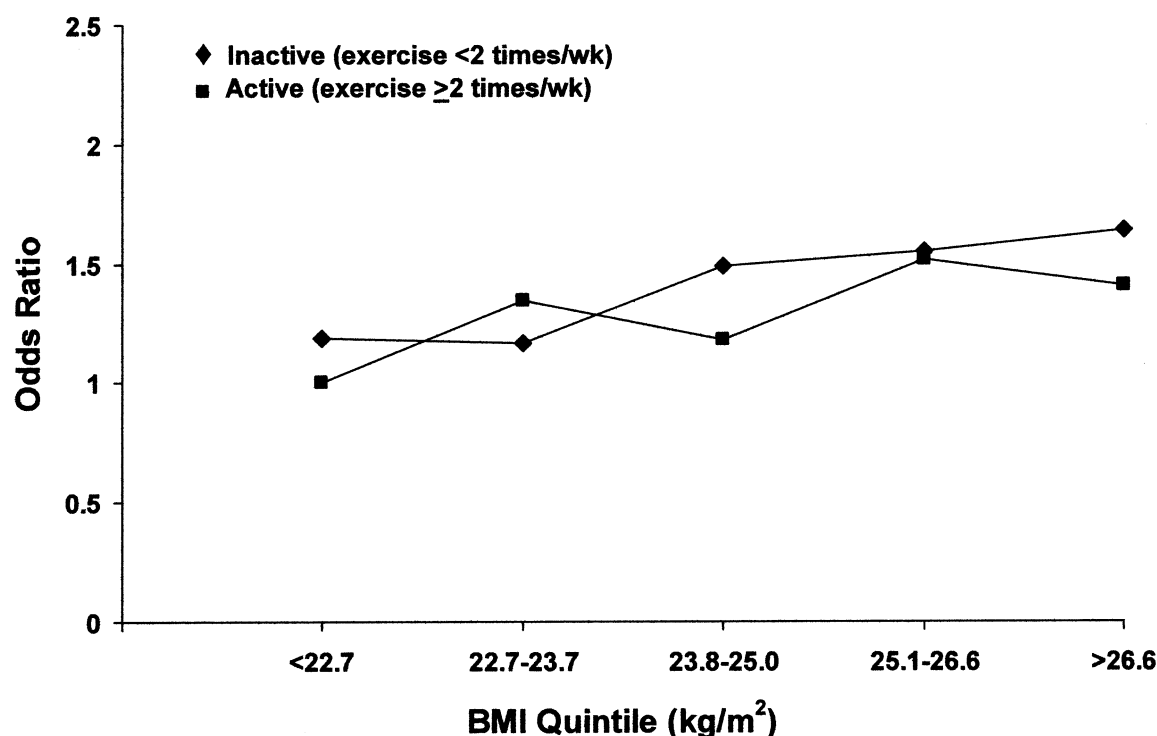


Fig 2. Age-adjusted ORs for decreased GFR (<60 mL/min/1.73 m² [<1.00 mL/s/1.73 m²]) at 14-year follow-up according to BMI quintile and physical activity level at baseline. Referent group (OR, 1.0): active with BMI less than 22.7 kg/m² (P for trend: inactive, $P = 0.004$; active, $P = 0.007$).

CKD across BMI quintiles (Fig 2). We did not find statistically significant effect modification of the association between BMI and CKD by baseline physical activity (exercise $\geq$ or < 2 times/wk; $P = 0.31$), diabetes status ($P = 0.46$), or age ($>$ or ≤ 60 years; $P = 0.51$).

After excluding participants who were obese at baseline (BMI ≥ 30 kg/m²; $n = 398$), BMI remained significantly associated with risk for CKD. In multivariable-adjusted analyses controlling for potential confounders, individuals in the highest BMI quintile had a 1.4-fold greater risk for developing a GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²) compared with those in the lowest quintile (OR, 1.44; 95% CI, 1.18 to 1.77; P for trend < 0.001).

In secondary analyses, we examined changes in BMI between baseline and 14-year follow-up. Overall, mean change in BMI was $+3.1\% \pm 7.6\%$ (SD), varying from $+0.99\% \pm 10.8\%$ in those who were obese at baseline to $+3.6\% \pm 7.1\%$ in those with a BMI less than 25.0 kg/m² at baseline ($P < 0.001$). Adjusting for potential confounders, men for whom BMI

increased by greater than 10% had a significantly greater risk for having a GFR less than 60 mL/min/1.73 m² (<1 mL/s/1.73 m²) at 14 years (OR, 1.27; 95% CI, 1.06 to 1.53), whereas men with either weight loss or lower levels of weight gain had nonsignificant increases in CKD risk compared with those for whom BMI remained overall stable (within 5% of their baseline level). Again, additional adjustment for potential mediators only slightly attenuated these associations (Table 4). In stratified analyses adjusting for both potential confounders and mediators, BMI increases greater than 10% were associated most strongly with CKD risk in men who were obese at baseline (OR, 2.70; 95% CI, 1.05 to 6.94 versus stable weight) compared with those who were overweight (OR, 1.15; 95% CI, 0.86 to 1.54) or normal weight (OR, 1.25; 95% CI, 0.97 to 1.61) at baseline (data not shown).

DISCUSSION

In this large prospective cohort study, we found that greater BMI was associated signifi-

Table 4. ORs and 95% CI for CKD at 14-Year Follow-Up According to 14-Year Changes in BMI

BMI Change (%)	No. of Patients	Age-Adjusted OR (95% CI)	Model 1* OR (95% CI)	Model 2† OR (95% CI)
< -5	1,246	1.12 (0.94-1.34)	1.17 (0.98-1.39)	1.09 (0.91-1.31)
-5 to +5	5,670	1.00	1.00	1.00
> +5 to +10	2,321	1.10 (0.94-1.29)	1.13 (0.96-1.33)	1.13 (0.96-1.33)
> +10	1,669	1.27 (1.06-1.52)	1.27 (1.06-1.53)	1.24 (1.03-1.50)
<i>P</i> for trend		0.11	0.14	0.09

NOTE. CKD defined as a GFR less than 60 mL/min/1.73 m² (<1.00 mL/s/1.73 m²).

*Model 1 adjusted for age at baseline (40 to 44, 45 to 49, 50 to 54, 55 to 59, 60 to 64, 65 to 69, and ≥70 years), smoking at baseline (never, past, current), alcohol consumption at baseline (≥1 drink/d, 2 to 6 drinks/wk, ≤1 drink/wk), exercise (none, 1 time/mo to 4 times/wk, ≥5 times/wk), and parental history of myocardial infarction before the age of 60 years.

†Model 2 adjusted for all variables in model 1 plus diabetes (history at baseline or during follow-up), hypertension at baseline or follow-up (defined as self-reported systolic blood pressure ≥140 mm Hg or diastolic blood pressure ≥90 mm Hg or antihypertensive medication use regardless of blood pressure), elevated cholesterol level at baseline or during follow-up (defined as ≥240 mg/dL [6.21 mmol/L] or history of past or current use of cholesterol-lowering medication), and development of cardiovascular disease during follow-up (defined as self-reported myocardial infarction, stroke, angina, coronary artery bypass graft, or percutaneous coronary angioplasty).

cantly with risk for CKD after an average follow-up of 14 years. This association was observed consistently across various measures of kidney function, including GFR (<60 mL/min/1.73 m² [<1 mL/s/1.73 m²]) and plasma creatinine level (≥1.5 mg/dL [≥133 μmol/L]). Even after controlling for potential biological mediators, baseline BMI remained significantly associated with CKD, suggesting that the risk incurred by a greater BMI may not be mediated solely by hypertension, diabetes, elevated cholesterol level, or the development of cardiovascular disease. Risk for CKD was consistently greatest for men who were hypertensive at baseline, across all BMI levels. Physical activity did not modify the association between BMI and risk for CKD.

Men who gained weight (>10% of their baseline BMI) during the 14-year follow-up had an approximately 30% significant increase in risk for CKD than men for whom BMI remained overall stable. Moreover, this effect of weight gain was strongest for men who were obese at baseline. Conversely, those who lost weight or had lower levels of weight gain had nonsignificant increases in CKD risk compared with those with stable weight, which possibly may be caused in part by unmeasured comorbid conditions associated with both weight loss and CKD.

Few studies evaluated BMI as a potential risk factor in the development of kidney disease, and available results are conflicting. BMI and central obesity were related to microalbuminuria in

some,^{37,38} but not all,³⁹ studies. Similarly, studies of BMI and CKD showed variable results, depending on study design and method of measuring kidney function.

In a cross-sectional analysis of data from the National Health and Nutrition Examination Survey, morbid obesity (defined as BMI ≥ 35 kg/m²) was related to CKD, but these associations were mediated by the effects of diabetes and hypertension.¹² Additional cross-sectional analyses suggested an association between BMI and elevated serum creatinine levels¹⁸ and between central fat distribution and decreased creatinine clearance.¹⁹ Conversely, other studies did not find such associations between BMI or waist circumference and CKD.^{15,20}

Results of our study are consistent with longitudinal data from Framingham, which suggested that baseline BMI predicts subsequent kidney disease after a mean follow-up of 18.5 years.¹³ In this community-based cohort of 2,585 men and women, each unit increase in BMI SD was associated with a 1.2-fold increased risk for new-onset kidney disease, controlling for age, sex, baseline GFR, smoking, and diabetes status.

Mechanisms by which BMI is related to kidney disease are uncertain. Obesity may promote kidney damage through both hemodynamic and hormonal effects.^{40,41} In studies of kidney pathology, glomerular injury was described in animal models of obesity in the absence of diabetes.⁴¹ Although similar pathobiological mechanisms appear to underlie both nephrosclerosis and ath-

erosclerosis,⁴² it is unclear whether obesity-related nephrosclerosis is related to the same microvascular injury seen in atherosclerosis.

Increasingly, evidence suggests that CKD and cardiovascular disease share common risk factors and are outcomes of the same predisposing disorders.^{6,10-17} Demonstration of common risk factors may indicate that cardiovascular and kidney diseases are mediated by common underlying mechanisms. The concept of shared risk factors is supported further by the identification of BMI as a risk factor for CKD, in addition to such known precursors as hypertension, diabetes, and hypercholesterolemia.

Strengths of our study include the large number of outcome events and participants, its prospective design, and the relative homogeneity of its participants, which reduces confounding by such variables as socioeconomic status and access to health care.

Several limitations should be considered. First, information on height and weight were self-reported; thus, random misclassification of body mass is possible. However, validation studies of other health professional cohorts have shown the reliability of self-reported information on height, weight, and cardiovascular risk factors.^{43,44} In addition, because of the prospective method of data collection, potential misclassification of BMI most likely would result in underestimation of the association between BMI and risk for CKD.

Second, interpretation of our findings also may be limited by potential misclassification of kidney function by using estimated GFR, such as the MDRD Study equation. Because muscle mass affects creatinine generation and all equations used to estimate GFR include creatinine level,^{6,35} the association of higher BMI with risk for CKD may be related in part to greater muscle mass in these individuals. However, even after excluding individuals with a baseline BMI of 30.0 kg/m² or greater, BMI remained significantly associated with risk for CKD. In addition, the association between BMI and CKD was strengthened when we used a more stringent criterion ($\text{GFR} < 50 \text{ mL/min/1.73 m}^2$ [$0.83 \text{ mL/s/1.73 m}^2$]) for CKD. Moreover, the MDRD Study equation has been shown to be an accurate predictor of GFR, in particular for GFR less than 60 mL/min/1.73 m² ($<1 \text{ mL/s/1.73 m}^2$), independent of body weight and height,^{35,45,46} and is recommended by the

National Kidney Foundation for estimation of GFR.⁶

Third, because creatinine measures were not available at baseline, we could not confirm that participants were free of CKD at the time of enrollment. Although physicians were excluded from the study if they reported a history of kidney disease, defined as renal failure or insufficiency, early stages of CKD may remain asymptomatic and undetected. However, examination of change in kidney function most likely would result in an underestimation of the association between BMI and GFR because of effects of BMI on GFR at baseline.

Fourth, despite adjustments for a large number of potential confounders of the association between BMI and CKD, residual confounding remains a possible alternative explanation for our findings given the study's observational design.

Finally, because all participants in the PHS were male health professionals and the vast majority (94.3%) were white, our results may not necessarily be extrapolated to other populations, including women. However, we have no a priori basis for believing that the biological mechanisms by which BMI may lead to CKD differ in other populations compared with the PHS cohort.

In conclusion, our study shows a significant association between BMI, as well as increases in BMI, and risk for CKD after 14 years. The increased risk was explained only partly by such potential mediators as hypertension, diabetes, high cholesterol level, and the development of cardiovascular disease before kidney function assessment. Potential health benefits of preventing overweight and obesity might include reduction in CKD risk.

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